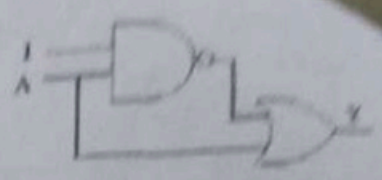


1. The output Y of the combinational circuit shown to the right is:

B. 1

C. A

D. $\bar{A}$



2- The range of signed decimal numbers that can be represented by 7 bits is:

A. -31 to +31

B. -63 to +63

C. -64 to +63

D. -127 to +127

3- The octal representation of $(156)_{10}$ is:

A. $(231)_8$

B. $(234)_8$

C. $(236)_8$

D. $(232.01)_8$

4- In 16-bit 2's complement representation, the decimal number -28 is:

A. 00000000000011100

C. 1111111111101100

B. 1111111111100100

D. 0000000000001100

5- The result of $(F0A00002)_{16} - (89D34FA9)_{16}$ is:

A. $(666CB0F9)_{16}$

B. $(70ACB0F9)_{16}$

C. $(766CB0F9)_{16}$

D. None of the above

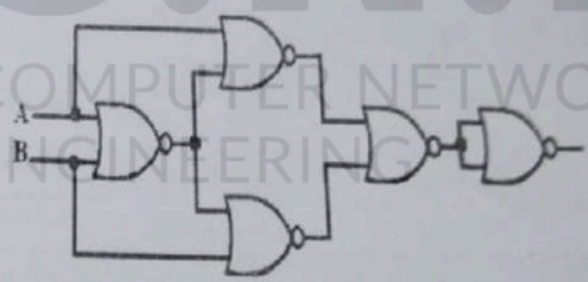
6- The combinational circuit shown in the right is:

A. $\bar{A}\bar{B} + AB$

B. $\bar{A} + \bar{B}$

C. $AB + A\bar{B}$

D. $A \cdot B$



7- The result of adding the hexadecimal numbers $(2C5AFF)_{16} + (1D7C67)_{16}$ is:

A. $(49D226)_{16}$

C. $(48C666)_{16}$

B. $(43E567)_{16}$

D. None of the above

8- Using 6-bit signed numbers (two's complement), which of the following decimal operations causes overflow?

A. +10

C. + -5

B. -20 + 5

D. -15 + -20

Q1:

1) $\overline{A+1} = \bar{A}$, $\bar{A}+A=1$ B.

2) $\frac{1}{64} \frac{0}{64} \frac{0}{64} \frac{0}{64} \frac{0}{64} \frac{0}{64} \frac{0}{64}$ - 64 C.

$\frac{0}{32} \frac{1}{16} \frac{1}{8} \frac{1}{4} \frac{1}{2} \frac{1}{1}$ + 63

3) $\begin{array}{r|l} 156 & 4 \\ 19 & 3 \\ 2 & 2 \\ 0 & \times \end{array}$ $\div 8$ carry $\uparrow$

$(234)_8$ B.

$\begin{array}{r} 2 \\ 8 \overline{) 19} \\ -16 \\ \hline 3 \end{array}$

$\begin{array}{r} 19 \\ 8 \overline{) 156} \\ -8 \\ \hline 76 \\ -72 \\ \hline 4 \end{array}$

4) $(28)_{10} \rightarrow (11100)_2$

16-bit representation:

$(0000000000001100)_2$

$\downarrow$ 2's complement:

$(11111111111100100)_2$

B.

$\div 2$ Carry $\uparrow$

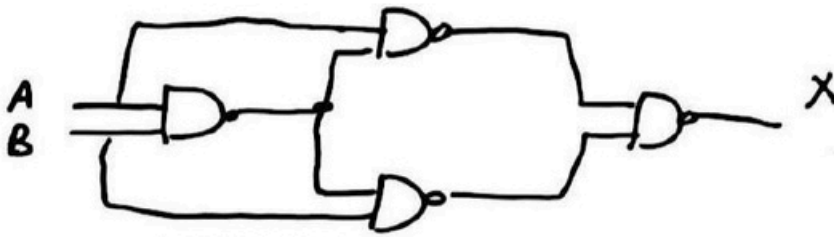
$\begin{array}{r|l} 28 & 0 \\ 14 & 0 \\ 7 & 1 \\ 3 & 1 \\ 1 & 1 \\ 0 & \times \end{array}$

5)

$$\begin{array}{r}
 E \quad F \quad \overset{25}{A} F F F F \quad 18 \\
 \cancel{F} \quad \cancel{0} \quad \cancel{A} \quad \cancel{0} \quad \cancel{0} \quad \cancel{0} \quad \cancel{2} \\
 \hline
 8 \quad 9 \quad D \quad 3 \quad 4 \quad F \quad A \quad 9 \quad - \\
 6 \quad 6 \quad C \quad C \quad B \quad 0 \quad 5 \quad 9
 \end{array}$$

Answer:

6)



$$X = \left[(\overline{A}B)(A) \quad (\overline{A}B)(B) \right]$$

Answer: C.

$$\begin{aligned}
 X &= (\overline{A}B)(A) + (\overline{A}B)(B) \\
 &= (\overline{A} + \overline{B})A + (\overline{A} + \overline{B})B \\
 &= \cancel{0} \quad \cancel{A} \quad \cancel{A} + A\overline{B} + \overline{A}B + \cancel{0} \quad \cancel{B} \quad \cancel{B} = \overline{A}B + A\overline{B}
 \end{aligned}$$

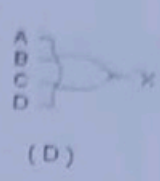
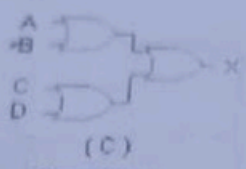
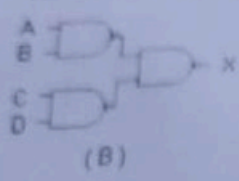
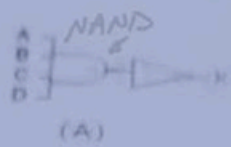
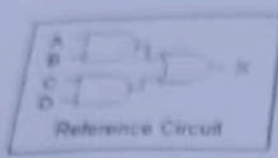
7)

$$\begin{array}{r}
 \overset{1}{2} \overset{1}{C} \overset{1}{5} \overset{1}{A} F F \\
 \hline
 1 D 7 C 6 7 + \\
 4 9 D 5 6 6
 \end{array}$$

$$\begin{aligned}
 15 + 7 &= 16 + 6 \\
 15 + 1 + 6 &= 16 + 6 \\
 10 + 12 + 1 &= 16 + 5 \\
 6 + 7 &= 13 \\
 12 + 13 &= 16 + 9
 \end{aligned}$$

Answer: D.

Each of the figures (A to D) is equivalent to the reference circuit



- A. (A)
- C. (C)

- B. (B)
- D. (D)

10. The equivalent binary representation of the following octal number $(701)_8$ is:
 A. $(111000101)_2$ B. $(101001001)_2$ C. $(111000001)_2$ D. $(011001001)_2$

11. If $F(x, y, z, w) = \pi(1, 2, 4, 8, 13, 14, 15)$, the complement of F ($\bar{F}$) is:

- A. $\pi(0, 3, 5, 6, 7, 9, 10, 11, 12)$
- B. $\sum(1, 2, 4, 8, 13, 14, 15)$
- C. $\sum(0, 3, 5, 6, 7, 9, 10, 11, 12)$
- D. $\pi(1, 2, 4, 8, 13, 14, 15)$

الوصات
 أي جواب ثاني علمه لا
 ممكن فيه don't care

12. The equivalent hexadecimal value of the binary number $(1001110101010)_2$ is:

- A. $(13AA)_{16}$
- B. $(9D50)_{16}$
- C. $(47AB)_{16}$
- D. $(9CA0)_{16}$



Problem#2: Find the most simplified SOP expression for

$$F(A, B, C, D) = \sum(1, 3, 5, 6, 7, 8, 10, 11, 12, 13, 14)$$

together with the don't care conditions $d(A, B, C, D) = \sum(0, 2, 4, 9)$.

You must show all steps including the drawing of the cells and grouping cells together, etc.
[5pts]

$\overline{AB} \backslash CD$	00	01	11	10
00	X	1	1	X
01	X	1	1	1
11	1	1	0	1
10	1	X	1	1

كتابة k-map
2 points

$$F = \overline{A} + \overline{B} + \overline{C} + \overline{D}$$

1 point

من الهمز يعرف :

$\overline{AB} \backslash CD$	00	01	11	10
00	X	1	1	X
01	X	1	1	1
11	1	1	0	1
10	1	X	1	1

grouping $\leadsto$ 2 points.

Problem # 3: The function $F(x, y, z, w) = \pi(0, 2, 4, 5, 9, 10, 13)$ is simplified to $F(x, y, z, w) = (x + w) \cdot (z + \bar{w}) \cdot (y + w)$ are there any don't care conditions? If yes, what are they? [3pts]

$xy \backslash zw$	00	01	11	10
00	0			0
01	0	0		
11		0		
10		0		0

رسم K-map
وتصديق على الأمتار
1.5 points

والتي ما استخدم K-map
طريقة آخر ← 0

نأخذ POS (الأمتار)

$$F = (x + w)(z + \bar{w})(y + w)$$

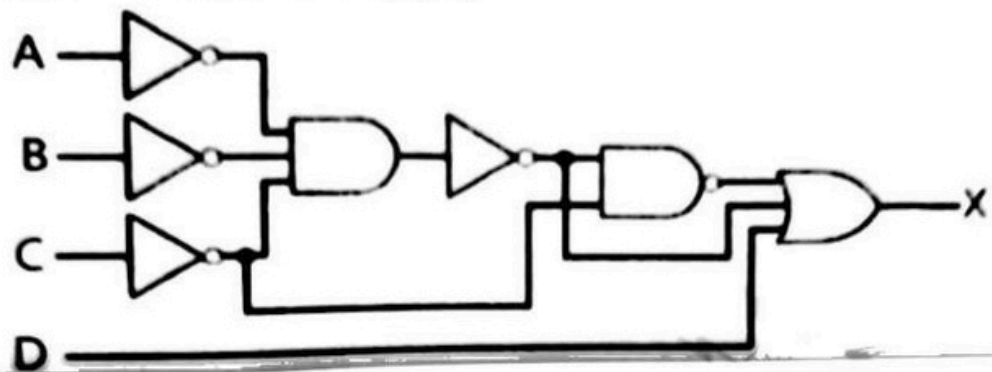
$xy \backslash zw$	00	01	11	10
00	0	X		0
01	0	0		X
11		0		
10	X	0		0

$$d = \Sigma(1, 6, 8) \rightarrow$$

(1.5 points)

don't care
علامه 0.5

Problem # 4: Based on the below combinational circuit, find the minimal form of X function using Boolean algebra rules. [5pts]



$$F = \overline{[\overline{A} \overline{B} \overline{C} \cdot \overline{C}]} + \overline{A} \overline{B} \overline{C} + D$$

2 points

$$z = \overline{A} \overline{B} \overline{C} : \text{نقطة}$$

$$F = \overline{\overline{A} \cdot \overline{C}} + \overline{A} + D$$

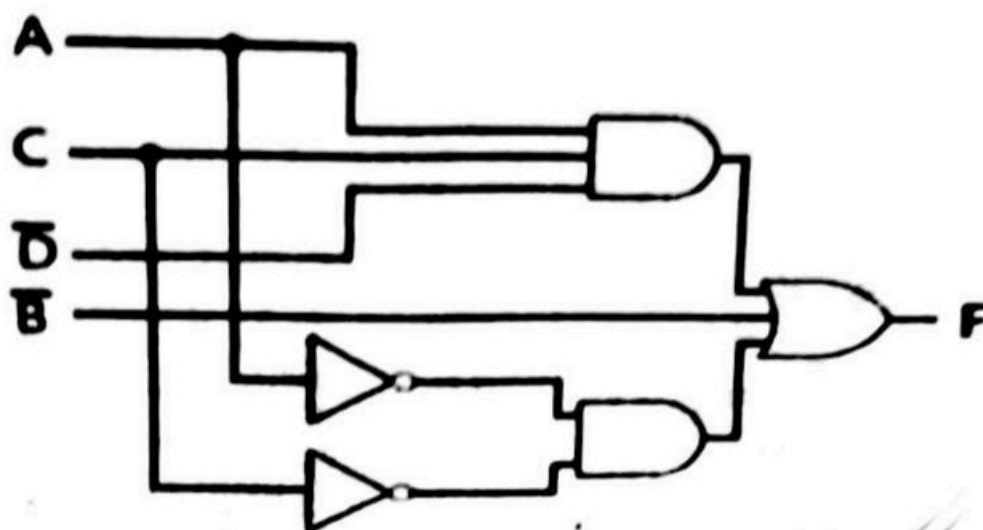
$$= z + C + \overline{A} + D = (\overline{A} + \overline{C}) + C + D$$

2 points

$$F = 1. \rightarrow \text{Final answer: 1 point.}$$



Problem # 5: Implement the below combinational circuit using only NAND gates. [5pts]



remember: using NAND only:

Not : A ——— D ———

AND : A ——— D ———
B ——— D ———

OR : A ——— D ———
B ——— D ———

Answer:

